HANZALA RASHID

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PROFESSIONAL SUMMARY

AI/ML Engineer with experience building multi-agent LLM systems, voice and text RAG pipelines, event-driven streaming, and computer vision applications. Specialized in full-stack ML microservices, local open-source LLM integrations (Ollama), and production deployments using Docker, Qdrant, and Linux. Proven track record in taking complex AI architectures from design to production.

TECHNICAL SKILLS

ML & LLM Agents:	Multi-Agent Systems, Open-Source LLMs (Ollama), RAG Pipelines, LangGraph, Qdrant, Parent-Child Chunking, QLoRA, Whisper STT
Frameworks & Tools:	Ollama, Streamlit, FastAPI, Next.js 14, TensorFlow, OpenCV, Scikit-Learn, LangChain, BGE-M3
Data & Streaming:	PostgreSQL, Debezium CDC, Redpanda (Kafka), Redis, Data Ingestion & ETL Pipelines
DevOps & Languages:	Docker, Docker Compose, Ubuntu Linux, Python, SQL, TypeScript, Bash

WORK EXPERIENCE

wAI Industries (incl. ISSM.AI) **Remote / On-site**

Junior ML & DevOps Developer *Aug 2026 – Present*

- Offered full-time role following internship to manage core ML integrations and infrastructure operations.
- Containerized Python microservices using Docker and Docker Compose for robust production environments.
- Prepared release builds, engineered custom patches, and deployed applications directly onto client Ubuntu Linux servers.
- Maintained day-to-day server updates and system health monitoring to ensure 99.9% uptime for client workloads.

Machine Learning Engineer Intern – ISSM.AI *Oct 2025 – Aug 2026*

- Developed FastAPI endpoints for real-time model inference, streaming response handlers, and asynchronous pipelines.
- Researched and benchmarked open-source model optimizations to enhance inference speed and output accuracy.
- Engineered custom web scrapers and unstructured data cleaning scripts to construct vectorizable datasets for RAG.
- Integrated bespoke LLM and computer vision capabilities into active client enterprise platforms.

KEY PROJECTS

OTTO – Multilingual & Multimodal Voice/Text RAG System **Next.js, FastAPI, LangGraph, Qdrant**

- Architected a voice/text RAG assistant supporting English, Urdu, and Roman Urdu using LangGraph and Next.js 14.
- Built an event-driven ingestion pipeline syncing PostgreSQL to Qdrant vector DB via Debezium CDC and Redpanda (Kafka).
- Implemented parent-child chunking with BGE-M3 embeddings, cross-encoder reranking, Whisper STT, and Edge-TTS.

MLL – Multi-Agent Open-Source LLM Game Suite **Streamlit, Ollama, LangChain**

- Created an interactive multi-agent UNO game suite utilizing Streamlit and a custom pythonic game engine rulebook.
- Configured local agents (Gemma, Mistral, Llama 3) with distinct personalities for real-time gameplay banter.
- Engineered prompt pipelines enabling synchronized multi-agent turn execution and live user-to-agent interaction.

Smart Housing Society 2.0 (Final Year Project) **Python, OpenCV, RAG, Web Portal**

- Built a resident portal with voting, ticketing, RAG query assistant, and ANPR license plate/facial recognition.

EDUCATION

DHA Suffa University **2022 – 2026**

Bachelor of Science in Data Science (8th Semester) *Karachi, Pakistan*

CERTIFICATIONS

• AI Engineer Core Track: LLM, RAG & Agents (Udemy)	**Mar 2026**	• Microsoft Certified: Azure AI Fundamentals (AI-900)	**Oct 2025**
• Claude Code in Action (Anthropic)	**Jul 2026**	• Intro to Model Context Protocol (MCP) (Anthropic)	**Jul 2026**